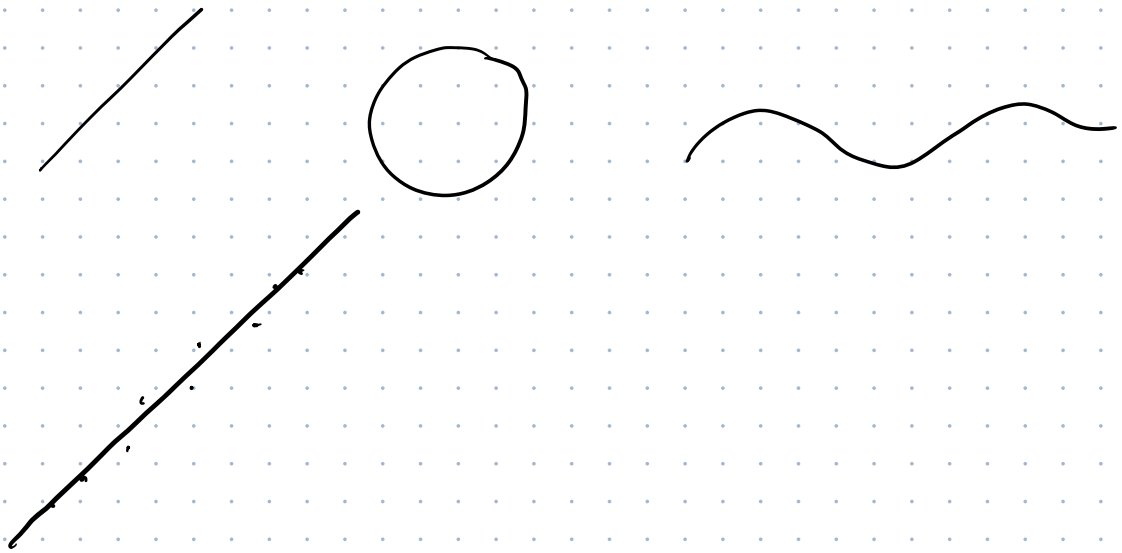
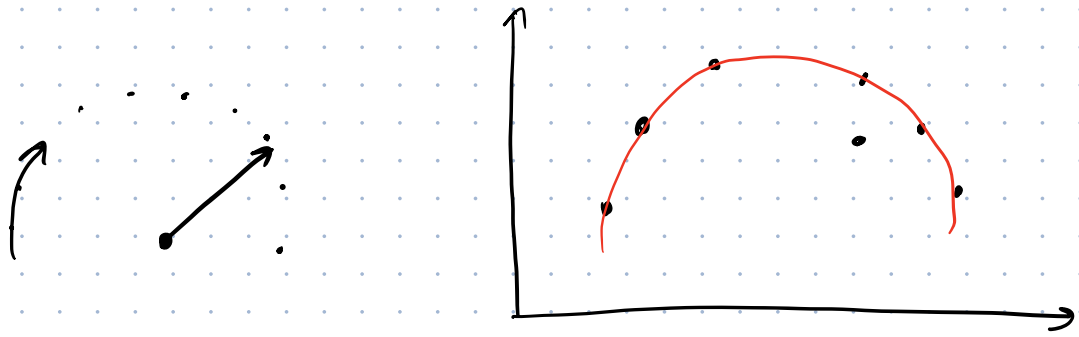
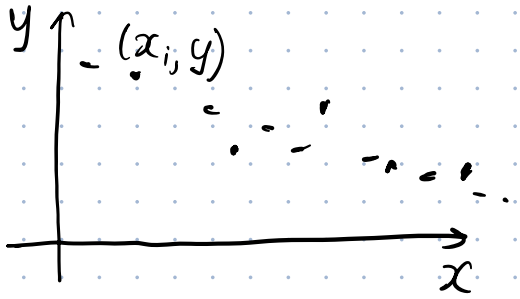
РАСПОЗНАВАНИЕ ОБРАЗОВ 03.03.26





МНК (МЕТОД НАИМЕНЬШИХ КВАДРАТОВ)



$y = ax + b$ ← МОДЕЛЬ

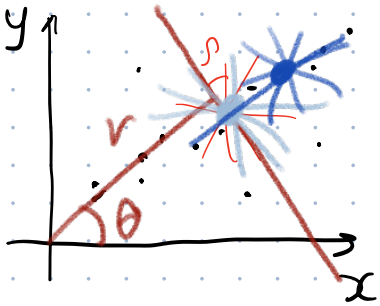
$\{(x_i, y_i)\}_{i=1}^{n}$

$$e_i = y_i - y_m(x_i) = y_i - ax_i - b$$

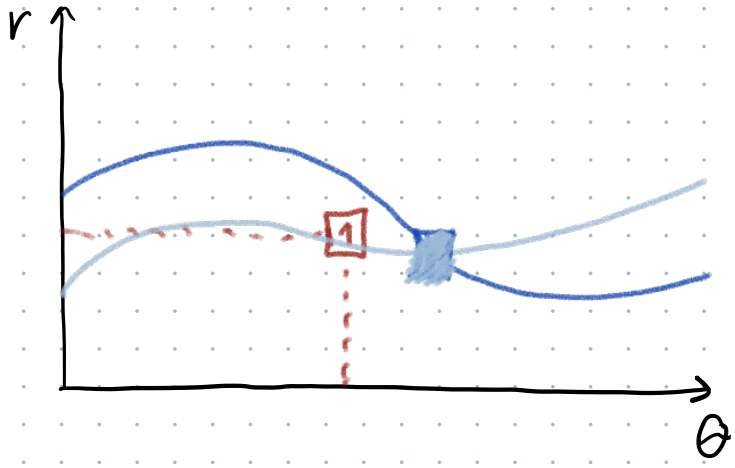
$$E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - ax_i - b)^2$$

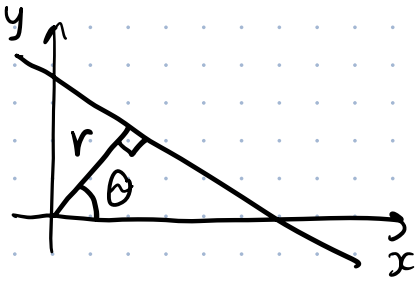
$$\begin{cases} \dfrac{\partial E}{\partial a} = 2\sum \ldots = 0 \\ \dfrac{\partial E}{\partial b} = 2\sum \ldots = 0 \end{cases} \Rightarrow \begin{matrix} \hat{a} = \ldots \\ \hat{b} = \ldots \end{matrix}$$

Hough transform (ПРЕОБРАЗОВАНИЕ ХАФА)

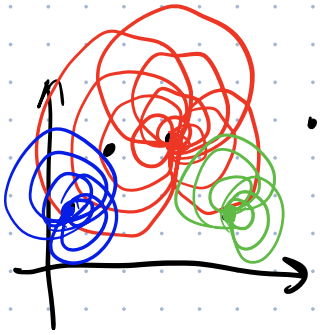


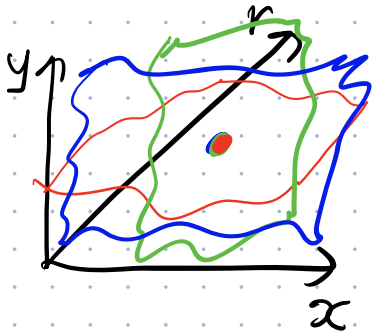
АККУМУЛЯТОР





ДЛЯ ОКР.:

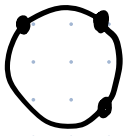




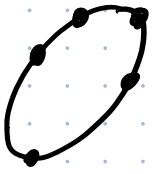
RANdom SAmple Consensus (RANSAC)

1) НАХ. СЛУЧ. ПОДВЫБ. ВХ. ДАННЫХ
2) СТРОИМ МОДЕЛЬ
3) ОЦЕН. КАЧ. МОД. ОТН. ДАННЫХ

2



3



5

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + 1 = 0$$

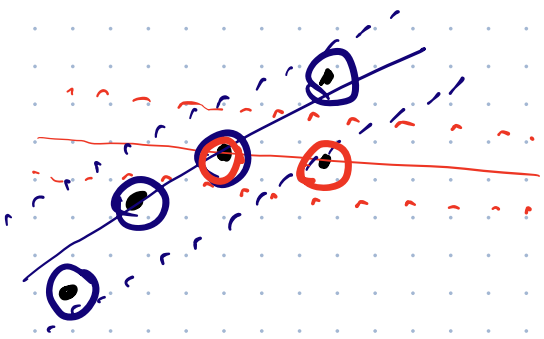
$$P = \begin{pmatrix} A \\ B \\ \dots \end{pmatrix}; \quad X_i = \begin{pmatrix} x_i^2 \\ x_i y_i \\ y_i^2 \\ x_i \\ y_i \end{pmatrix}$$

$$P^T X_i = -1 = X_i^T P$$

$$i \in [1, 5] \cap \mathbb{N}$$

$$\begin{pmatrix} X_1^T \\ X_2^T \\ \dots \\ X_5^T \end{pmatrix} P = \begin{pmatrix} -1 \\ \dots \\ -1 \end{pmatrix}$$

ОЦЕНКА КАЧ. МОДЕЛИ



inliers

$$\text{score}(\text{data}, \text{model}) = \sum_{i=1}^{n} \mathbb{I}(\rho(\text{data}, \text{model}) \le th)$$

K - КОЛ-ВО ТОЧЕК НА 1 МОДЕЛЬ

α - ДОЛЯ ИНЛАЕРОВ

P - ЖЕЛАЕМАЯ ВЕР-ТЬ УСПЕХА

m - ЧИСЛО ИТЕРАЦИЙ

$$(1 - \alpha^K)^m = 1 - P \quad | \ln(\ )$$

$$m = \left\lceil \frac{\ln(1-P)}{\ln(1-\alpha^K)} \right\rceil$$